

ImPRESS — Time-to-Event Proportional-Hazards Assessment

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1 Introduction

This document provides a pre-unblinding assessment of the **proportional-hazards (PH) assumption** for the Cox models reported in the time-to-event section of the main mock-up report (`make_rdtte.R`), in **54 subjects** (cohort 2). It addresses consistency-review point **C3** (SAP states the PH assumption is assessed by standard diagnostics; the main report shows hazard ratios without the diagnostics).

For each endpoint the Cox model is `coxph(Surv(aval, 1 - cnsr) ~ trtcd)`, where `trtcd` is the long-term planned dose (0/25/50/100 mg) — the same specification used for the reported hazard ratios. The PH assumption is assessed with:

1. **Scaled Schoenfeld residual test** (`survival::cox.zph`) — a global test plus a per-term test; small p-values indicate a time-varying (non-proportional) effect.
2. **Schoenfeld residual plots** — scaled residuals against transformed time; a non-zero slope signals departure from PH.
3. **Complementary log-log survival plots** — $\log(-\log S(t))$ against $\log(\text{time})$ by dose; approximately parallel curves are consistent with PH.

All analyses use sham-randomised data and are descriptive.

Interpretation guide. Treat the global Schoenfeld p-value as the headline: $p > 0.10$ — no evidence against PH; $0.05\text{--}0.10$ — borderline, inspect plots; $p < 0.05$ — evidence of non-proportionality, consider a time-varying coefficient, stratification, or reporting a time-averaged hazard ratio with this caveat.

2 Per-endpoint diagnostics

2.1 Overall survival (OS)

34 events in 54 subjects.

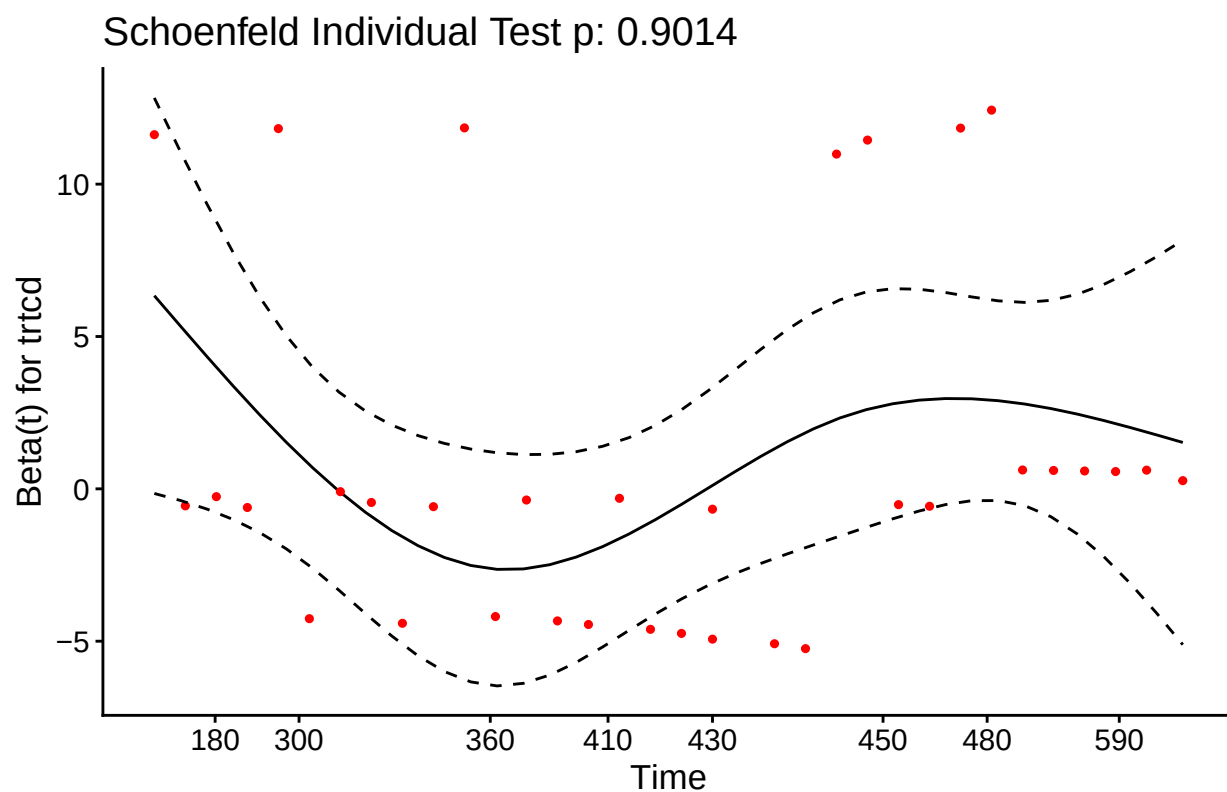
Table 1: Scaled Schoenfeld test — Overall survival

	Term	Chi-square	df	p-value
trtcd	trtcd	0.578	3	0.9014
GLOBAL	GLOBAL (overall)	0.578	3	0.9014

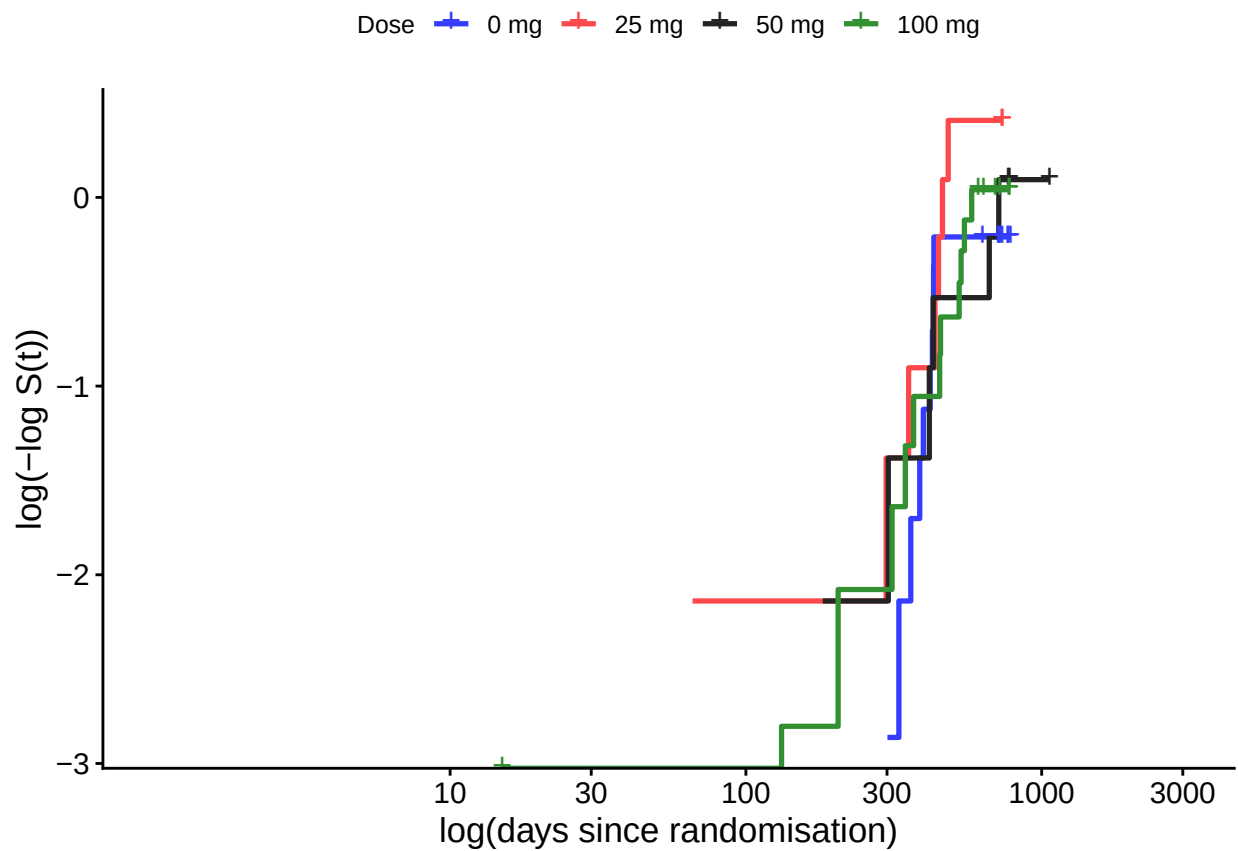
Global Schoenfeld p-value: 0.901 — No evidence against PH.

2.1.1 Scaled Schoenfeld residuals

Global Schoenfeld Test p: 0.9014



2.1.2 Complementary log-log survival



2.2 Overall survival at 12 months (OS12M)

12 events in 54 subjects.

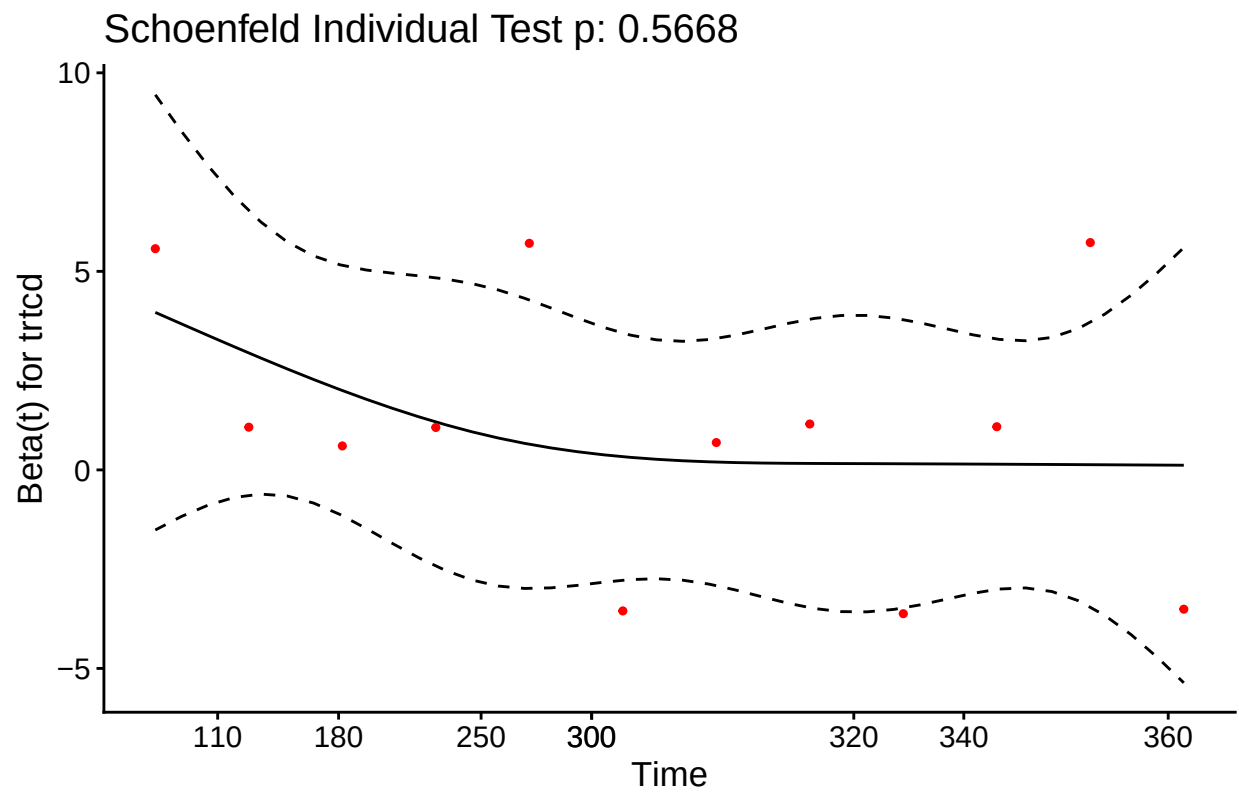
Table 2: Scaled Schoenfeld test — Overall survival at 12 months

	Term	Chi-square	df	p-value
trtcd	trtcd	2.027	3	0.5668
GLOBAL	GLOBAL (overall)	2.027	3	0.5668

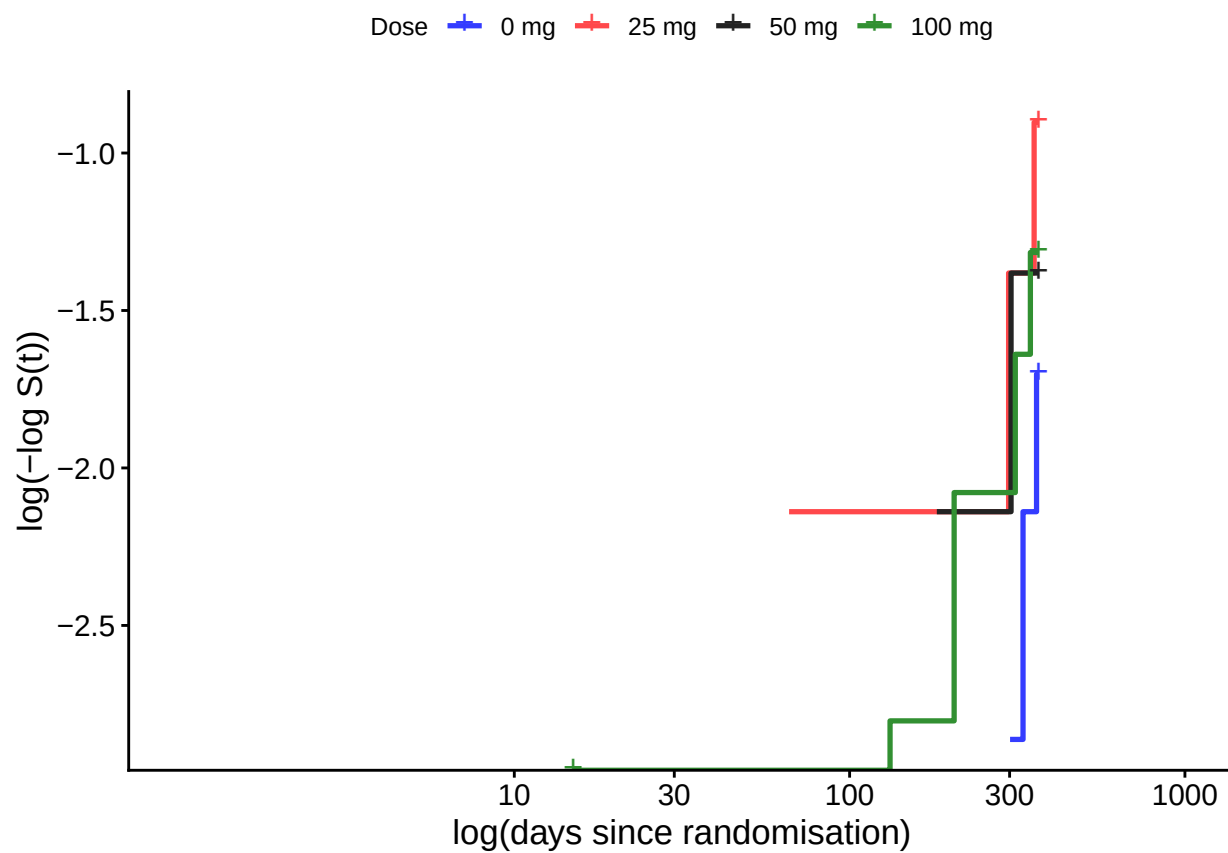
Global Schoenfeld p-value: 0.567 — No evidence against PH.

2.2.1 Scaled Schoenfeld residuals

Global Schoenfeld Test p: 0.5668



2.2.2 Complementary log-log survival



2.3 Overall survival at 24 months (OS24M)

34 events in 54 subjects.

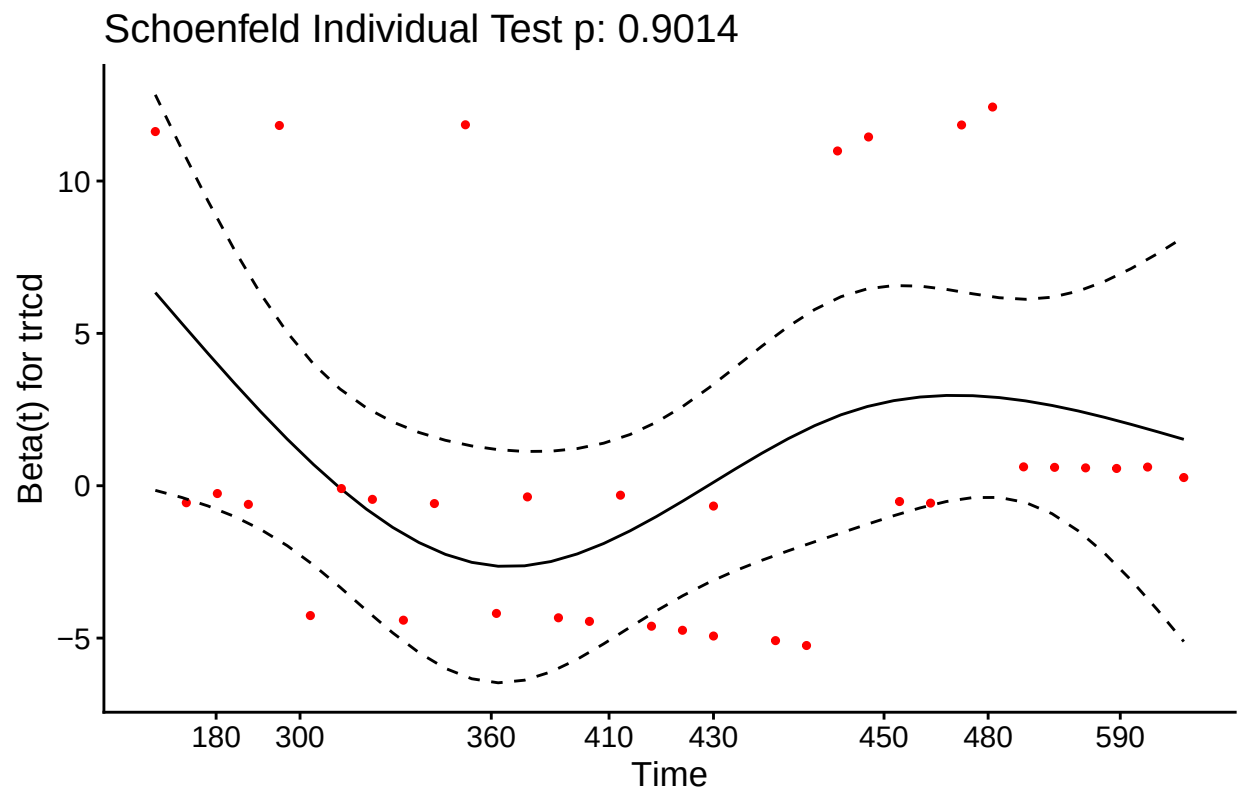
Table 3: Scaled Schoenfeld test — Overall survival at 24 months

	Term	Chi-square	df	p-value
trtcd	trtcd	0.578	3	0.9014
GLOBAL	GLOBAL (overall)	0.578	3	0.9014

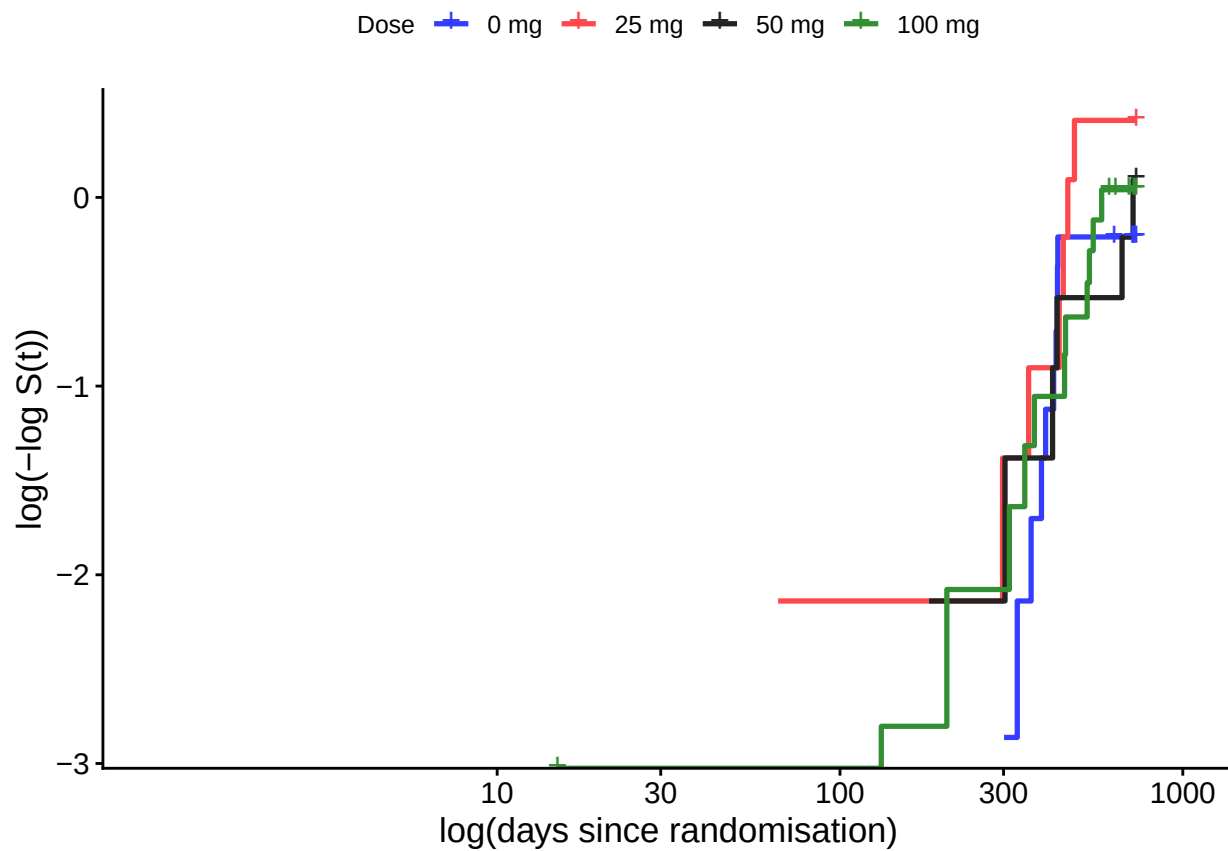
Global Schoenfeld p-value: 0.901 — No evidence against PH.

2.3.1 Scaled Schoenfeld residuals

Global Schoenfeld Test p: 0.9014



2.3.2 Complementary log-log survival



2.4 Progression-free survival (PFS)

47 events in 54 subjects.

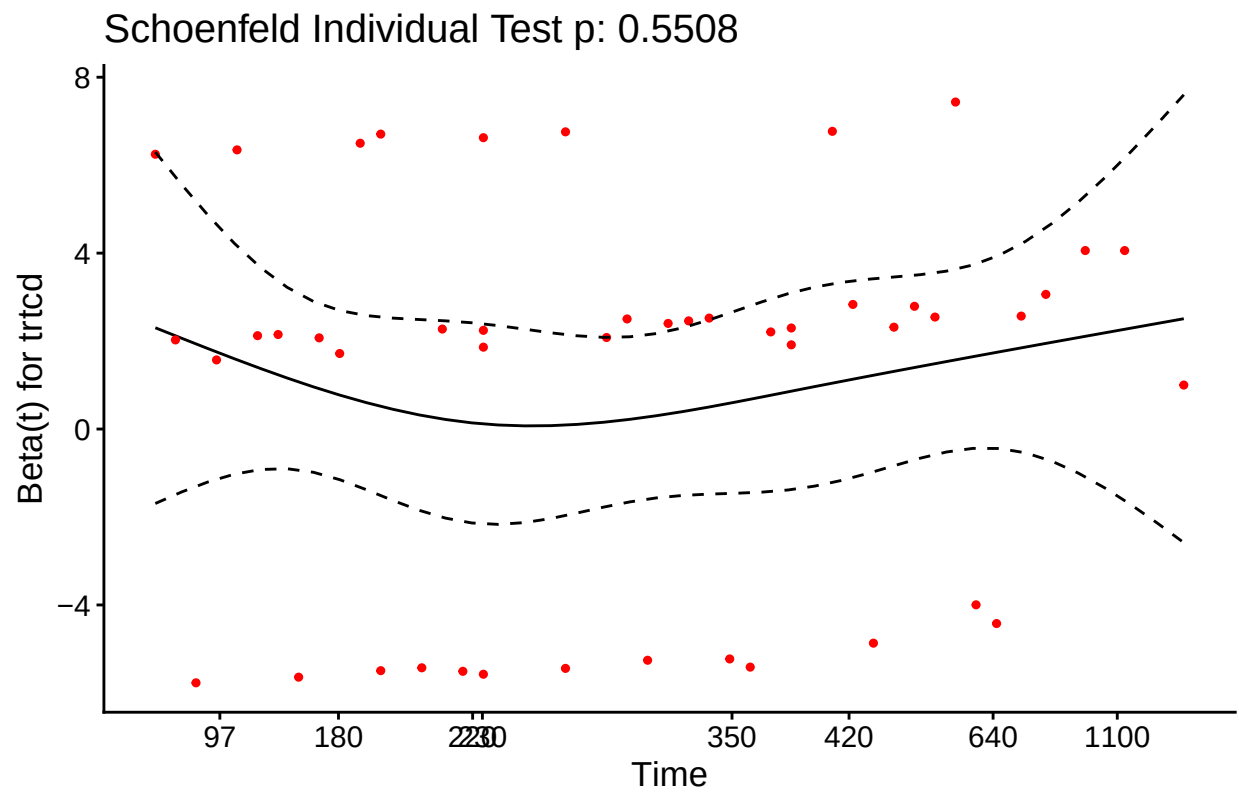
Table 4: Scaled Schoenfeld test — Progression-free survival

	Term	Chi-square	df	p-value
trtcd	trtcd	2.106	3	0.5508
GLOBAL	GLOBAL (overall)	2.106	3	0.5508

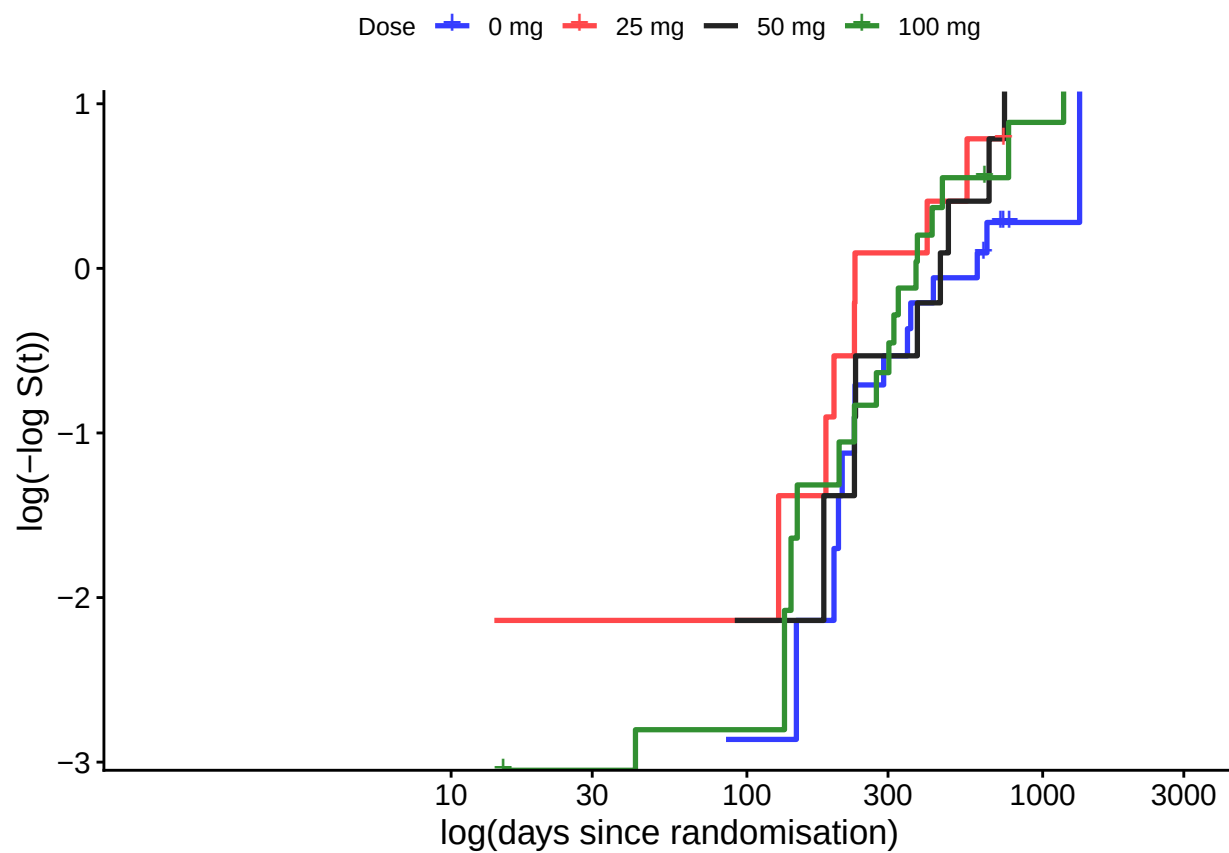
Global Schoenfeld p-value: 0.551 — No evidence against PH.

2.4.1 Scaled Schoenfeld residuals

Global Schoenfeld Test p: 0.5508



2.4.2 Complementary log-log survival



2.5 6-month progression-free survival (PFS6M)

9 events in 54 subjects.

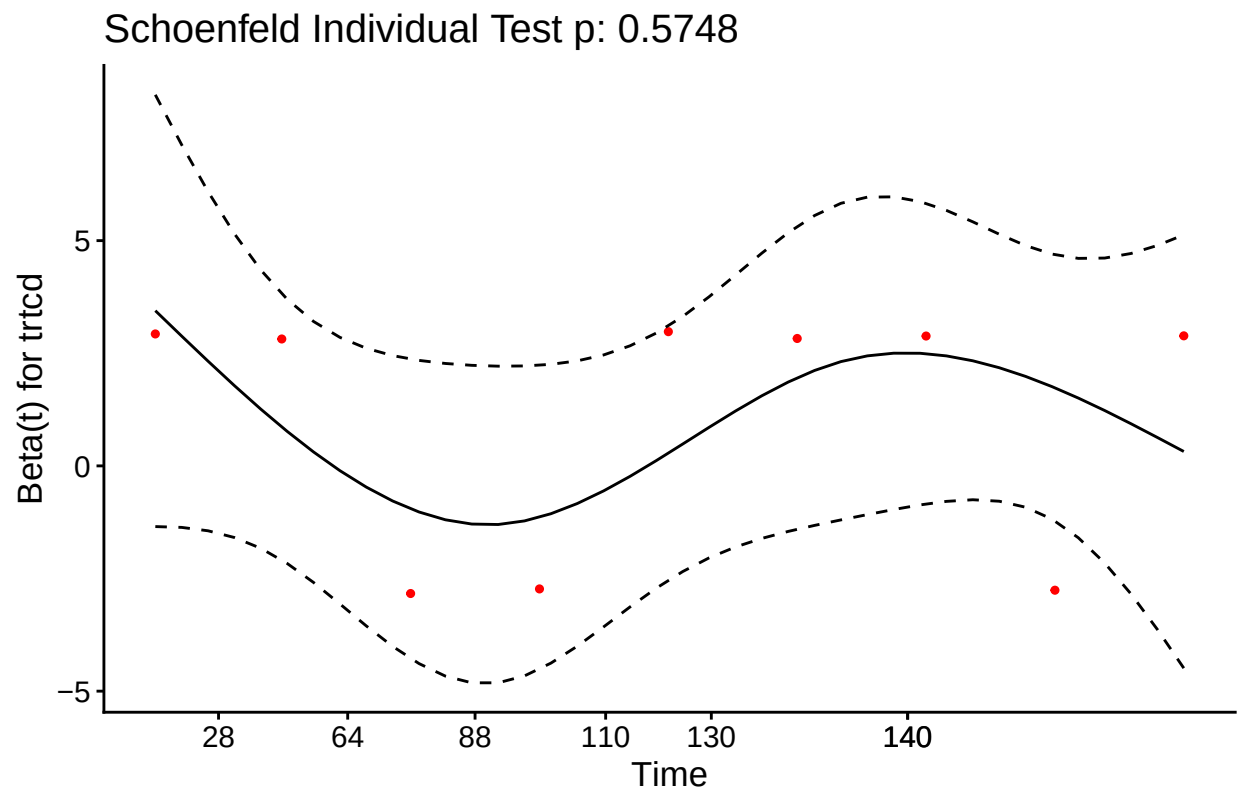
Table 5: Scaled Schoenfeld test — 6-month progression-free survival

	Term	Chi-square	df	p-value
trted	trted	1.988	3	0.5748
GLOBAL	GLOBAL (overall)	1.988	3	0.5748

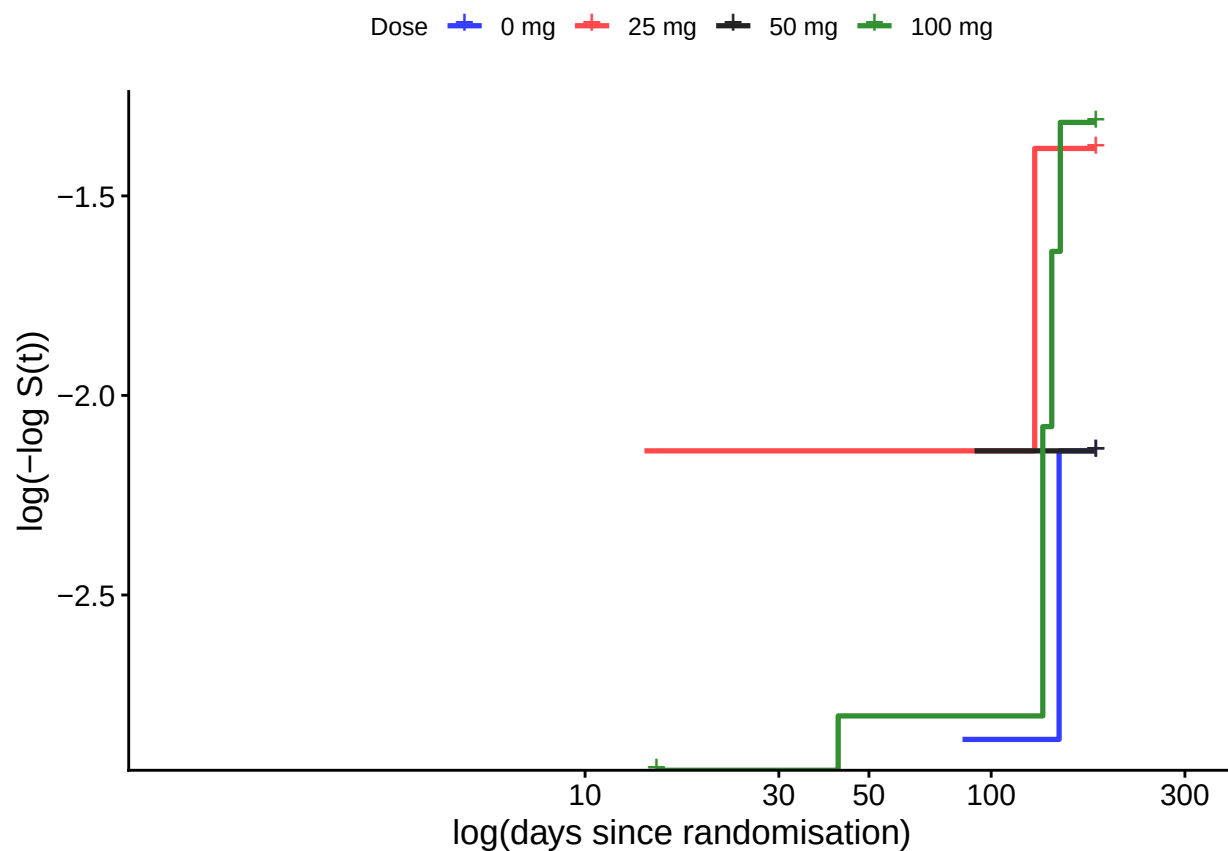
Global Schoenfeld p-value: 0.575 — No evidence against PH.

2.5.1 Scaled Schoenfeld residuals

Global Schoenfeld Test p: 0.5748



2.5.2 Complementary log-log survival



3 Summary

Table 6: Global proportional-hazards test by endpoint

PARAMCD	Parameter	Global Schoenfeld p	Conclusion
OS	Overall survival	0.9014	No evidence against PH
OS12M	Overall survival at 12 months	0.5668	No evidence against PH
OS24M	Overall survival at 24 months	0.9014	No evidence against PH
PFS	Progression-free survival	0.5508	No evidence against PH
PFS6M	6-month progression-free survival	0.5748	No evidence against PH

Endpoints flagged as borderline or non-proportional should be interpreted with the corresponding plots; for those a time-averaged hazard ratio is reported in the main document with this caveat, and a time-varying-coefficient or stratified sensitivity analysis can be added at database lock.

Generated from `reports/model_fit/impress_tte_ph_check.Rmd` — sham-randomised data.